1 Fundamentals: Logic, sets, numbers and vectors

1.1 Real Numbers, Eulidean Spaces

Axioms of Addition:

• **Axioms:**

A1 (Associativity)	$x + (y + z) = (x + y) + z$	$\forall x, y, z \in \mathbb{R}$
A2 (Neutral element)	$x + 0 = x$	$\forall x \in \mathbb{R}$
A3 (Inverse element)	$x + (-x) = 0$	$\forall x \in \mathbb{R}\ \exists(-x) \in \mathbb{R}$
A4 (Commutativity)	$x + y = y + x$	$\forall x, y \in \mathbb{R}$

Axioms of Multiplication:

• **Axioms:**

M1 (Associativity)	$x \cdot (y \cdot z) = (x \cdot y) \cdot z$	$\forall x, y, z \in \mathbb{R}$
M2 (Neutral element)	$x \cdot 1 = x$	$\forall x \in \mathbb{R}$
M3 (Inverse element)	$x \cdot x^{-1} = 1$	$\forall x \in \mathbb{R} \setminus \{0\}\ \exists x^{-1} \in \mathbb{R}$
M4 (Commutativity)	$x \cdot z = z \cdot x$	$\forall x, z \in \mathbb{R}$

Distributivity (Compatibility of addition and mult.):

• **Axiom:**

$$\textbf{D} \qquad x \cdot (y + z) = xy + xz = (y + z) \cdot x \qquad \forall x, y, z \in \mathbb{R}$$

Order Axioms:

• **Axioms:**

O1 (Reflexivity)	$x \leq x$	$\forall x \in \mathbb{R}$
O2 (Transitivity)	$(x \leq y \wedge y \leq z) \Rightarrow x \leq z$	
O3 (Antisymmetry)	$(x \leq y \wedge y \leq x) \Rightarrow x = y$	
O4 (Totality)	$\forall x, y \in \mathbb{R} :\ x \leq y$ or $y \leq x$	

Compatibility (Consistency with addition and mult.):

• **Axioms:**

K1	$x \leq y \Rightarrow x + z \leq y + z$	$\forall x, y, z \in \mathbb{R}$
K2	$x \geq 0,\ y \geq 0 \Rightarrow xy \geq 0$	$\forall x, y \in \mathbb{R}$

R 1.1.5:

• The set $\mathbb{Q}$ of rational numbers, equipped with addition, multiplication, and the order relation $\leq$, satisfies the above axioms.

V (Completeness axiom):

• The set $\mathbb{R}$ is **order complete**: For any two nonempty sets $A, B \subset \mathbb{R}$ such that

$$a \leq b \quad \text{for all } a \in A,\ b \in B,$$

there exists a number $c \in \mathbb{R}$ satisfying

$$a \leq c \leq b \quad \forall a \in A,\ b \in B.$$

C 1.1.6:

• **The consequences of above axioms are:**

1. Uniqueness of the additive and multiplicative inverse.
2. $0 \cdot x = 0 \quad \forall x \in \mathbb{R}$.
3. $(-1) \cdot x = -x \quad \forall x \in \mathbb{R}$, in particular $(-1)^2 = 1$.
4. $y \geq 0 \iff (-y) \leq 0$.
5. $y^2 \geq 0 \quad \forall y \in \mathbb{R}$, in particular $1 = 1 \cdot 1 \geq 0$.
6. If $x \leq y$ and $u \leq v$, then $x + u \leq y + v$.
7. If $0 \leq x \leq y$ and $0 \leq u$, then $xu \leq yu$.

T (Completeness of the real numbers):

• $\mathbb{R}$ is a commutative ordered field that is order-complete.

C 1.1.7 (Archimedean Principle):

• 1. For $x \in \mathbb{R}$ and $y > 0$ there exists $n \in \mathbb{N}$ such that

$$ny > x.$$

2. For every $\varepsilon > 0$ there exists $n \in \mathbb{N}$ such that

$$\frac{1}{n} < \varepsilon.$$

T 1.1.8:

• For every $t \geq 0$, $t \in \mathbb{R}$, the equation $x^2 = t$ has a solution in $\mathbb{R}$.

D (Bounded set - upper/lower bounded):

• A set $A \in \mathbb{R}$ is called upper bounded if

$$\exists b \in R \quad \forall a \in A : a \leq b.$$

Each such b is called an upper bound for A. (Similarly for lower bound.)

T (Supremum Infimum):

• 1. Maximum and minimum (if exists) are unique

2. Every non-empty subset bounded below (above) has a unique infimum (supremum).

3. $S = \sup(X) \Leftrightarrow$

$$(\forall x \in X : x \leq S) \wedge (\forall \varepsilon > 0\ \exists x \in X : \quad x > S - \varepsilon)$$

4. $I = \inf(X) \Leftrightarrow$

$$(\forall x \in X : x \geq I) \wedge (\forall \varepsilon > 0\ \exists x \in X : \quad x < I + \varepsilon)$$

D (Maximum):

• Maximum is:

$$\max\{x, y\} := \begin{cases} x, & \text{if } y \leq x, \\ y, & \text{if } x \leq y. \end{cases}$$

D (Minimum):

• Minimum is:

$$\min\{x, y\} := \begin{cases} y, & \text{if } y \leq x, \\ x, & \text{if } x \leq y. \end{cases}$$

T (Properties of the absolute value):

• Properties:

1. $|x| \geq 0$.
2. $|x + y| \leq |x| + |y| \qquad$ (triangle inequality).
3. $|x + y| \geq \big||x| - |y|\big|$ (inv. triangle inequality).

T (Young's inequality):

• For every $\varepsilon > 0$ and all $x, y \in \mathbb{R}$ it holds:

$$2|xy| \leq \varepsilon x^2 + \frac{1}{\varepsilon} y^2.$$

1.2 Vectors and Vector Spaces

D (inner product):

• For $x, y \in \mathbb{R}^n$ the standard inner product is

$$x \cdot y = \langle x, y \rangle = \sum_{i=1}^{n} x_i y_i$$

D (Euclidean norm):

• The Euclidean norm is

$$\|x\| = \sqrt{\sum_{i=1}^{n} x_i^2}$$

D (Euclidean distance):

• The Euclidean distance is

$$d(x, y) = \sqrt{\sum_{i=1}^{n} (x_i - y_i)^2}$$

D (Triangle inequality):

• For all $x, y, z \in \mathbb{R}^n$:

$$\|x - z\| \leq \|x - y\| + \|y - z\|$$

1.3 Complex Numbers

D (Complex number):

- A complex number is

$$z = a + ib, \quad a, b \in \mathbb{R}$$

D (set of complex numbers):

- The set of complex numbers is

$$\mathbb{C} = \{a + ib \mid a, b \in \mathbb{R}\}$$

D (Terminology of complex numbers):

- Terminology
  - $a = \mathrm{Re}(z)$ — real part
  - $b = \mathrm{Im}(z)$ — imaginary part
  - $i^2 = -1$
  - If $a = 0$, the number is purely imaginary

D (Complex numbers Addition and Subtraction):

- Let $z_1 = x_1 + iy_1$, $z_2 = x_2 + iy_2$.

$$z_1 + z_2 = (x_1 + x_2) + i(y_1 + y_2)$$

$$z_1 - z_2 = (x_1 - x_2) + i(y_1 - y_2)$$

D (Complex numbers Multiplication):

- Let $z_1 = x_1 + iy_1$, $z_2 = x_2 + iy_2$.

$$z_1 z_2 = x_1 x_2 - y_1 y_2 + i(x_1 y_2 + y_1 x_2)$$

D (Complex conjugate):

- For a complex number $z = x + iy$, the complex conjugate of z is given by

$$\overline{z} = \overline{x + iy} = x - iy.$$

D (Complex numbers Division):

- To divide numbers we use conjugate:

$$\frac{z}{w} = \frac{z\overline{w}}{w\overline{w}} = \frac{z\overline{w}}{|w|^2}$$

D (Properties of the Conjugate):

- Properties are:
  - $z\overline{z} = |z|^2$
  - $z + \overline{z} = 2\,\mathrm{Re}(z), \quad z - \overline{z} = 2i\,\mathrm{Im}(z)$
  - $\overline{\overline{z}} = z$
  - $\overline{z \pm w} = \overline{z} \pm \overline{w}$
  - $\overline{zw} = \overline{z}\,\overline{w}$
  - $z = \overline{z}$, if $z \in \mathbb{R}$
  - $\overline{z} = -z$, z is purely imaginary

D (Modulus):

- The magnitude $|z|$ of a complex number $z = a + ib$ is the length of the corresponding vector.

$$|z| = \sqrt{a^2 + b^2}$$

The distance between two complex numbers z1 and z2 is

$$d(z_1, z_2) = |z_2 - z_1|$$

Triangle inequality works also for complex numbers:

$$|z + w| \leq |z| + |w|$$

D (Euler's Formula):

- The exponential function can be represented by a series expansion as

$$e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}$$

If we substitute $x = it$ into this formula, we obtain Euler's formula.

$$e^{it} = \cos t + i \sin t$$

We can write $sin(t)$ and $cos(t)$ using complex exponential functions.

$$\cos(t) = \frac{e^{it} + e^{-it}}{2}$$

$$\sin(t) = \frac{e^{it} - e^{-it}}{2i}$$

D (Exponential with Complex Argument):

- We can also allow complex arguments for the exponential function. Then the following rules apply:
  - $e^{z+w} = e^z e^w$
  - $e^{a+ib} = e^a(\cos b + i \sin b)$
  - $e^{z+i2\pi} = e^z$

D (Polar Form):

- A complex number can also be described by specifying the distance from the origin and by specifying a suitable angle (**polar form**): $z = re^{i\varphi}$.
  - $r = |z| \geq 0$ - distance
  - $\varphi \in (-\pi, \pi]$ - polar angle
  - $\varphi = \arg(z) = \mathrm{arc}(z)$

D (Polar to Cartesian conversion):

- Polar to Cartesian conversion is:
  - $x = r\cos\varphi$
  - $y = r\sin\varphi$

D (Cartesian to Polar conversion):

- Cartesian to Polar conversion is:
  - $r = \sqrt{x^2 + y^2}$
  - $\varphi = \arctan\frac{y}{x}$

If $x = 0$:

$$\arg(iy) = \begin{cases} \frac{\pi}{2}, & y > 0 \\ -\frac{\pi}{2}, & y < 0 \end{cases}$$

D (Powers of Complex Numbers):

- The following rules must be observed:
  - $z_1 z_2 = r_1 r_2 e^{i(\varphi_1 + \varphi_2)}$
  - $\frac{z_1}{z_2} = \frac{r_1}{r_2} e^{i(\varphi_1 - \varphi_2)}$
  - $z^n = r^n e^{in\varphi}$

D (Roots from complex numbers):

- For $n \in \mathbb{N}$, the solutions to the equation $z^n = re^{i\varphi}$ are

$$z_k = r^{1/n} e^{i(\varphi/n + 2\pi k/n)}, \quad k = 0, \ldots, n-1$$

D (Quadratic Equations):

- For the quadratic equation

$$az^2 + bz + c = 0,$$

in complex numbers the solution is still

$$z_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

D (Fundamental Theorem of Algebra):

- Each polynom

$$p(z) = \sum_{k=0}^{n} a_k z^k, \quad a_k \in \mathbb{C}$$

can be factored into linear factors, i.e. written as

$$p(z) = a_n(z - z_1)(z - z_2)\ldots(z - z_n)$$

The numbers z_k are therefore precisely the zero points of $p(z)$ (with **multiplicity**).

C (non-real roots appear in conjugate pairs):

- Non-real roots of a polynomial with **real** coefficients occur in conjugate pairs $z, \overline{z}$.

2 Sequences

D (Sequence):

• A sequence $(a_n)_{n\in\mathbb{N}_0}$ (also written $(a_n)_{n\geq 0}$ or $(a_n)_{n=0}^{\infty}$) is an infinite list of numbers, where each entry is called a *term* of the sequence. A sequence can also be viewed as a function

$$f : \mathbb{N}_0 \to \mathbb{R} \quad (\text{or } f : \mathbb{N} \to \mathbb{R}),$$

whose values are $f(n) = a_n$.

D (Ways to Describe a Sequence):

• One way to describe a sequence is by giving an explicit formula for each term:

$$a_n = f(n).$$

Another possibility is to define the sequence recursively, constructing each new term from previous ones:

$$a_n = g(a_{n-1}, a_{n-2}, \dots).$$

The functions f and g are called the *rule of formation* of the sequence.

D (Convergence/Divergence):

• A sequence $(a_n)_{n\in\mathbb{N}_0} \subset \mathbb{R}$ is called *convergent* if there exists a number $L \in \mathbb{R}$ such that

$$\forall \varepsilon > 0\ \exists N > 0 \text{ such that } \forall n > N : |a_n - L| < \varepsilon.$$

The number L is called the *limit* of the sequence, and we write

$$\lim_{n\to\infty} a_n = L.$$

If no such L exists, the sequence is called *divergent*.

T (Uniqueness of the Limit):

• A convergent sequence has exactly one unique limit.

D (Divergence to infinity):

• Let $(a_n)_{n\in\mathbb{N}_0}$ be a sequence.

- If
  $$\forall M > 0\ \exists N > 0 \text{ such that } \forall n > N : a_n > M,$$
  then the sequence diverges to infinity and we write
  $$\lim_{n\to\infty} a_n = \infty.$$
- If
  $$\forall M > 0\ \exists N > 0 \text{ such that } \forall n > N : a_n < -M,$$
  then the sequence diverges to minus infinity and we write
  $$\lim_{n\to\infty} a_n = -\infty.$$

D (Subsequence):

• Let $(a_n)_{n\in\mathbb{N}_0}$ be a sequence in $\mathbb{R}$. A *subsequence* is a sequence of the form

$$(a_{n_k})_{k\in\mathbb{N}_0},$$

where $(n_k)_{k\in\mathbb{N}_0}$ is a sequence of non-negative integers satisfying

$$n_{k+1} > n_k \quad \text{for all } k \in \mathbb{N}_0.$$

L (Limit of a sequence and its subsequence):

• Let $(a_n)_{n\in\mathbb{N}_0}$ be a convergent sequence with limit $L \in \mathbb{R}$. Then every subsequence $(a_{n_k})_{k\in\mathbb{N}_0}$ also converges to the same limit L.

D (Accumulation Point):

• Let $(a_n)_{n\in\mathbb{N}_0}$ be a sequence in $\mathbb{R}$. A number $A \in \mathbb{R}$ is called an accumulation point of the sequence if

$$\forall \varepsilon > 0\ \forall N \in \mathbb{N}_0\ \exists n \geq N \text{ such that } |a_n - A| < \varepsilon.$$

Intuitively, every interval $(A - \varepsilon, A + \varepsilon)$ contains terms a_n for infinitely many indices n.

T (Characterization of Accumulation Points):

• Let $(a_n)_{n\in\mathbb{N}_0}$ be a sequence in $\mathbb{R}$. A number $A \in \mathbb{R}$ is an accumulation point of the sequence if and only if there exists a convergent subsequence $(a_{n_k})_{k\in\mathbb{N}_0}$ such that

$$\lim_{k\to\infty} a_{n_k} = A.$$

C (Accumulation Point of a Convergent Sequence):

• Every convergent sequence $(a_n)_{n\in\mathbb{N}_0}$ in $\mathbb{R}$ has exactly one accumulation point, which coincides with its limit.

C (Infinite Occurrence Near an Accumulation Point):

• Let A be an accumulation point of the sequence $(a_n)_{n\in\mathbb{N}_0}$. Then for every $\varepsilon > 0$ there exist infinitely many terms of the sequence lying in the interval $(A - \varepsilon, A + \varepsilon)$.

T (Algebra of Limits):

• Assume

$$\lim_{n\to\infty} a_n = K, \qquad \lim_{n\to\infty} b_n = L$$

with $K, L \neq \pm\infty$ and let C be a constant. Then

$$\lim_{n\to\infty} (a_n + b_n) = K + L$$
$$\lim_{n\to\infty} (a_n - b_n) = K - L$$
$$\lim_{n\to\infty} (a_n b_n) = KL$$
$$\lim_{n\to\infty} (Ca_n) = CK$$

If $L \neq 0$ and $b_n \neq 0$ then

$$\lim_{n\to\infty} \frac{a_n}{b_n} = \frac{K}{L}.$$

If $K < L$ then there exists N such that $a_n < b_n$ for all $n \geq N$. If there exists N such that $a_n \leq b_n$ for all $n \geq N$, then $K \leq L$.

T (Important Limits):

•

$$\lim_{n\to\infty} \frac{\log n}{n} = 0 = \lim_{n\to\infty} \frac{\ln n}{n}$$
$$\lim_{n\to\infty} n^{1/n} = 1$$
$$\lim_{n\to\infty} x^{1/n} = 1 \quad (x > 0)$$
$$\forall x \in \mathbb{R} : \quad \lim_{n\to\infty} \left(1 + \frac{x}{n}\right)^n = e^x$$
$$\forall x \in \mathbb{R} : \quad \lim_{n\to\infty} \frac{x^n}{n!} = 0$$

T (Sandwich Theorem):

• Let $(a_n)_{n\in\mathbb{N}_0}$ and $(c_n)_{n\in\mathbb{N}_0}$ be convergent sequences with

$$\lim_{n\to\infty} a_n = L, \qquad \lim_{n\to\infty} c_n = L$$

and suppose there exists N_0 such that

$$a_n \leq b_n \leq c_n \quad \text{for all } n \geq N_0.$$

Then $(b_n)_{n\in\mathbb{N}_0}$ is also convergent and

$$\lim_{n\to\infty} b_n = L.$$

D (Bounded Sequence):

• A sequence $(a_n)_{n\in\mathbb{N}_0}$ is called bounded below or bounded above if the set

$$M = \{a_n \mid n \in \mathbb{N}_0\}$$

has a lower bound or an upper bound respectively. A sequence that is bounded both above and below is called a bounded sequence.

D (Monotone Sequence):

• A sequence $(a_n)_{n\in\mathbb{N}_0}$ is called monotone decreasing if

$$m > n \Rightarrow a_m \leq a_n$$

and monotone increasing if

$$m > n \Rightarrow a_m \geq a_n.$$

It is called strictly monotone decreasing if

$$m > n \Rightarrow a_m < a_n$$

and strictly monotone increasing if

$$m > n \Rightarrow a_m > a_n.$$

T (Monotone Convergence Theorem):

• Every bounded and monotone sequence converges.

L (Convergent sequence = bounded):

• Every convergent sequence is bounded.

T Monotone Convergence:

• Let $(a_n)_{n\in\mathbb{N}_0}$ be a sequence of real numbers.

(i) A monotone sequence converges if and only if it is bounded.

(ii) If (a_n) is monotonically increasing, then

$$\lim_{n\to\infty} a_n = \sup\{a_n \mid n \in \mathbb{N}_0\}.$$

(iii) If (a_n) is monotonically decreasing, then

$$\lim_{n\to\infty} a_n = \inf\{a_n \mid n \in \mathbb{N}_0\}.$$

D Limit Superior and Limit Inferior:

• Let $(a_n)_{n\in\mathbb{N}_0}$ be a sequence.

- *limit superior* of the sequence

$$\limsup_{n\to\infty} a_n = \lim_{n\to\infty} \sup\{a_k \mid k \geq n\}$$

- *limit inferior* of the sequence.

$$\liminf_{n\to\infty} a_n = \lim_{n\to\infty} \inf\{a_k \mid k \geq n\}$$

L Properties of limsup and liminf:

• Let $(a_n)_{n\in\mathbb{N}_0}$ be a sequence. Then:

(i) $\limsup_{n\to\infty} a_n = \inf_{n\in\mathbb{N}} \left(\sup\{a_k \mid k \geq n\}\right)$

(ii) $\liminf_{n\to\infty} a_n = \sup_{n\in\mathbb{N}} \left(\inf\{a_k \mid k \geq n\}\right)$

(iii) If (a_n) is convergent, then

$$\lim_{n\to\infty} a_n = \limsup_{n\to\infty} a_n = \liminf_{n\to\infty} a_n.$$

(iv) A bounded sequence (a_n) converges if and only if

$$\lim_{n\to\infty} a_n = \limsup_{n\to\infty} a_n = \liminf_{n\to\infty} a_n.$$

T Characterization via limsup:

• Let $(a_n)_{n\in\mathbb{N}_0}$ be a bounded sequence and define

$$A = \limsup_{n\to\infty} a_n.$$

Then:

(i) A is an accumulation point of the sequence.

(ii) For every $\varepsilon > 0$, there are only finitely many indices n such that

$$a_n > A + \varepsilon.$$

(iii) For every $\varepsilon > 0$, there are infinitely many indices n such that

$$A - \varepsilon < a_n < A + \varepsilon.$$

An analogous statement holds for the limit inferior.

C Subsequence of a Bounded Sequence:

• Every bounded sequence of real numbers has at least one accumulation point and admits a convergent subsequence.

D Cauchy Sequence:

• A sequence $(a_n)_{n\in\mathbb{N}_0} \subset \mathbb{R}$ is called a *Cauchy sequence* if

$$\forall \varepsilon > 0\ \exists N \in \mathbb{N} \text{ such that } \forall m, n \geq N: \ |a_n - a_m| < \varepsilon.$$

L Cauchy Sequences are Bounded:

• Every Cauchy sequence is bounded.

T Cauchy Criterion:

• A sequence of real numbers converges if and only if it is a Cauchy sequence.

3 Series

D Series:

• A (formal) expression of the form

$$a_0 + a_1 + a_2 + \cdots = \sum_{n=0}^{\infty} a_n$$

is called a *series*. The terms a_n are called elements or summands. If the sequence of partial sums

$$s_n := \sum_{k=0}^{n} a_k$$

converges to a limit $L \in \mathbb{R}$, then the series is said to *converge*, and we write

$$\sum_{n=0}^{\infty} a_n = L.$$

Otherwise, the series is called *divergent*.

T Necessary Condition for Convergence:

• If the series $\sum_{n=0}^{\infty} a_n$ converges, then necessarily

$$\lim_{n\to\infty} a_n = 0.$$

L Linearity of Series:

• Let $\sum_{n=0}^{\infty} a_n$ and $\sum_{n=0}^{\infty} b_n$ be convergent series. Then:

$$\sum_{n=0}^{\infty}(a_n + b_n) = \sum_{n=0}^{\infty} a_n + \sum_{n=0}^{\infty} b_n,$$

$$\sum_{n=0}^{\infty} Ca_n = C\sum_{n=0}^{\infty} a_n \quad (C \in \mathbb{R}).$$

L Tail of a Series:

• Let $\sum_{n=0}^{\infty} a_n$ be a series. For any $N \in \mathbb{N}_0$:

$$\sum_{n=0}^{\infty} a_n \text{ converges} \quad \Longleftrightarrow \quad \sum_{n=N}^{\infty} a_n \text{ converges},$$

and in that case

$$\sum_{n=0}^{\infty} a_n = \sum_{n=0}^{N-1} a_n + \sum_{n=N}^{\infty} a_n.$$

T Series with Non-Negative Terms:

• Let $a_n \geq 0$ for all n. Then the sequence of partial sums

$$s_n = \sum_{k=0}^{n} a_k$$

is monotonically increasing. If (s_n) is bounded, then the series $\sum_{n=0}^{\infty} a_n$ converges; otherwise, it diverges.

T Comparison Test:

• Let $\sum a_n$, $\sum b_n$, $\sum c_n$ be series with non-negative terms and suppose that for some N:

$$c_n \leq b_n \leq a_n \quad \text{for all } n \geq N.$$

Then:

- If $\sum a_n$ converges, then $\sum b_n$ converges (majorant test).
- If $\sum c_n$ diverges, then $\sum b_n$ diverges (minorant test).

D Absolute and Conditional Convergence:

• A series $\sum a_n$ is called *absolutely convergent* if $\sum |a_n|$ converges.

• It is called *conditionally convergent* if $\sum a_n$ converges but $\sum |a_n|$ does not.

T Riemann Rearrangement Theorem:

• Let $\sum a_n$ be a conditionally convergent series of real numbers, and let $L \in \mathbb{R}$. Then there exists a bijection $\varphi : \mathbb{N}_0 \to \mathbb{N}_0$ such that

$$\sum_{n=0}^{\infty} a_{\varphi(n)} = L.$$

T Rearrangement of Absolutely Convergent Series:

• Let $\sum a_n$ be absolutely convergent and let $\varphi : \mathbb{N}_0 \to \mathbb{N}_0$ be a bijection. Then

$$\sum_{n=0}^{\infty} a_{\varphi(n)}$$

converges absolutely and

$$\sum_{n=0}^{\infty} a_n = \sum_{n=0}^{\infty} a_{\varphi(n)}.$$

T Leibniz Criterion:

• Let (a_n) be a monotonically decreasing sequence of non-negative real numbers with

$$\lim_{n\to\infty} a_n = 0.$$

Then the alternating series

$$\sum_{n=0}^{\infty} (-1)^n a_n$$

converges. Moreover, for all $m \in \mathbb{N}_0$:

$$\sum_{n=0}^{2m+1} (-1)^n a_n \leq \sum_{n=0}^{\infty} (-1)^n a_n \leq \sum_{n=0}^{2m} (-1)^n a_n.$$

T Cauchy Criterion for Series:

• The series $\sum_{n=0}^{\infty} a_n$ converges if and only if for every $\varepsilon > 0$ there exists N such that for all $n \geq m \geq N$:

$$\left| \sum_{k=m+1}^{n} a_k \right| \leq \varepsilon.$$

T Absolute Convergence Implies Convergence:

• Every absolutely convergent series converges, and

$$\left| \sum_{n=0}^{\infty} a_n \right| \leq \sum_{n=0}^{\infty} |a_n|.$$

T Root Test:

• Let (a_n) be a sequence and define

$$\rho = \limsup_{n\to\infty} |a_n|^{1/n}.$$

Then:

- If $\rho < 1$, the series $\sum a_n$ converges absolutely.
- If $\rho > 1$, the series $\sum a_n$ diverges.

T Ratio Test:

• Let (a_n) with $a_n \neq 0$ and define

$$\rho = \lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right|.$$

Then:

- If $\rho < 1$, the series $\sum a_n$ converges absolutely.
- If $\rho > 1$, the series $\sum a_n$ diverges.

T Cauchy Product:

• Let $\sum a_n$ and $\sum b_n$ be absolutely convergent. Then

$$\left(\sum_{n=0}^{\infty} a_n\right)\left(\sum_{n=0}^{\infty} b_n\right) = \sum_{n=0}^{\infty}\left(\sum_{k=0}^{n} a_{n-k} b_k\right),$$

and the series on the right converges absolutely.

D Power Series:

• A series of the form

$$\sum_{k=0}^{\infty} c_k (x-a)^k$$

is called a *power series* centered at a with coefficients c_k.

D Convergence of a Power Series:

• For fixed x, if the sequence of partial sums

$$s_n(x) = \sum_{k=0}^{n} c_k (x-a)^k$$

converges, then the power series is said to converge at x.

T Radius of Convergence:

• Every power series has a radius of convergence R such that:

- The series converges for $|x-a| < R$,
- The series diverges for $|x-a| > R$.

The interval $(a-R, a+R)$ is called the interval of convergence. The radius R is given by:

$$\rho = \limsup_{n\to\infty} |c_n|^{1/n}.$$

Then

$$R = \begin{cases} 0 & \text{if } \rho = \infty, \\ \frac{1}{\rho} & \text{if } 0 < \rho < \infty, \\ \infty & \text{if } \rho = 0. \end{cases}$$

4 Functions

4.1 Function basics

D Function:

- A function (mapping) is a rule that assigns to each admissible input exactly one output in the target set. The set of all possible inputs is called the *domain*, denoted by domain(f) or D. The set of all actually attained outputs is called the *image*, denoted by image(f) or W.

$$f : \text{domain}(f) \to \text{range}(f), \quad x \mapsto f(x).$$

The input is called the *independent variable*, the output the *dependent variable*.

D Restriction of a Function:

- Let $f : D(f) \to \mathbb{R}$ and let $D_0 \subseteq D(f)$. The restriction of f to D_0 is the function

$$f|_{D_0} : D_0 \to \mathbb{R}, \quad f|_{D_0}(x) = f(x).$$

Note that f and $f|_{D_0}$ are, in general, different functions.

D Bounded Functions:

- Let $f : D(f) \to \mathbb{R}$.
  - f is *bounded above* if there exists $M \in \mathbb{R}$ such that
  $$f(x) \le M \quad \forall x \in D(f).$$
  - f is *bounded below* if there exists $M \in \mathbb{R}$ such that
  $$f(x) \ge M \quad \forall x \in D(f).$$
  - f is *bounded* if there exists $M \in \mathbb{R}$ such that
  $$|f(x)| \le M \quad \forall x \in D(f).$$

D Injective, Surjective, Bijective:

- Let $f : X \to Y$.
  - f is *injective* if
  $$f(x_1) = f(x_2) \Rightarrow x_1 = x_2.$$
  - f is *surjective* if
  $$\forall y \in Y \ \exists x \in X : f(x) = y.$$
  - f is *bijective* if it is both injective and surjective.

D Composition:

- Let $f : X \to Y$ and $g : Y \to Z$. The composition is defined by

$$g \circ f : X \to Z, \quad (g \circ f)(x) = g(f(x)).$$

D Identity and Inverse:

- For a set X, the identity function is defined by

$$\text{id}_X(x) = x \quad \forall x \in X.$$

If $f : X \to Y$ is bijective, then there exists a unique function

$$f^{-1} : Y \to X$$

such that

$$f^{-1} \circ f = \text{id}_X, \quad f \circ f^{-1} = \text{id}_Y.$$

This function is called the *inverse* of f.

D Monotonicity:

- Let $f : D(f) \subseteq \mathbb{R} \to \mathbb{R}$.
  - f is *monotonically increasing* if
  $$x_1 < x_2 \Rightarrow f(x_1) \le f(x_2).$$
  - f is *strictly increasing* if
  $$x_1 < x_2 \Rightarrow f(x_1) < f(x_2).$$
  - f is *monotonically decreasing* if
  $$x_1 < x_2 \Rightarrow f(x_1) \ge f(x_2).$$
  - f is *strictly decreasing* if
  $$x_1 < x_2 \Rightarrow f(x_1) > f(x_2).$$

T Strict Monotonicity Implies Injectivity:

- Every strictly monotone function is injective.

D Even and Odd Functions:

- Let $f : \mathbb{R} \to \mathbb{R}$.
  - f is *even* if
  $$f(-x) = f(x).$$
  - f is *odd* if
  $$f(-x) = -f(x).$$

D Convexity and Concavity (Geometric):

- The graph of a function is called
  - *convex* if it bends to the left when moving from left to right,
  - *concave* if it bends to the right when moving from left to right.

4.2 Limits of Functions

D Limit of a Function:

- Let $f : D(f) \to \mathbb{R}$ and $x_0 \in \mathbb{R}$. A number $L \in \mathbb{R}$ is called the limit of $f(x)$ at x_0 if

$$\forall \varepsilon > 0 \ \exists \delta > 0 \text{ such that } \quad x \in D(f), \ |x - x_0| < \delta \Rightarrow |f(x) - L| < \varepsilon.$$

The limit is uniquely determined.

D One-Sided Limits:

- One-Sided Limits
  - Right-hand limit:
  $$\lim_{x \to x_0^+} f(x) = L$$
  - Left-hand limit:
  $$\lim_{x \to x_0^-} f(x) = L$$

D Limits at Infinity:

- Limits at Infinity
  - $\lim_{x \to \infty} f(x) = L$
  - $\lim_{x \to -\infty} f(x) = L$

D Infinite Limits:

- Infinite Limits
  - $\lim_{x \to c} f(x) = \infty$
  - $\lim_{x \to c} f(x) = -\infty$

T Limit Laws:

- Let

$$\lim_{x \to c} f(x) = K, \quad \lim_{x \to c} g(x) = L, \quad K, L \ne \pm\infty.$$

Then:

  - $\lim_{x \to c}(f + g) = K + L$
  - $\lim_{x \to c}(f - g) = K - L$
  - $\lim_{x \to c}(fg) = KL$
  - $\lim_{x \to c}(Af) = AK$
  - If $L \ne 0$: $\lim_{x \to c} \frac{f}{g} = \frac{K}{L}$

4.3 Continuity

D Continuity:

• Let $f : D \to \mathbb{R}$ and $x_0 \in D$. The function f is called *continuous at* x_0 if

$$\forall \varepsilon > 0 \ \exists \delta > 0 \text{ such that } |x - x_0| < \delta \ \Rightarrow \ |f(x) - f(x_0)| < \varepsilon.$$

The function is called *continuous* if it is continuous at every point in its domain.

D One-Sided Continuity:

• Let $f : D(f) \to \mathbb{R}$ and $x_0 \in D(f)$.

- f is *right-continuous* at x_0 if

$$\lim_{x \to x_0^+} f(x) = f(x_0).$$

- f is *left-continuous* at x_0 if

$$\lim_{x \to x_0^-} f(x) = f(x_0).$$

T Sequential Characterization of Continuity:

• Let $f : D \subseteq \mathbb{R} \to \mathbb{R}$ and $x_0 \in D$. Then f is continuous at x_0 if and only if for every sequence (x_n) with $x_n \to x_0$ we have

$$f(x_n) \to f(x_0).$$

T Algebra of Continuous Functions:

• Let $f, g : D \subseteq \mathbb{R} \to \mathbb{R}$ be continuous. Then:

- $f + g$ is continuous
- $f - g$ is continuous
- cf is continuous for any constant $c \in \mathbb{R}$
- $f \cdot g$ is continuous
- $\frac{f}{g}$ is continuous if $g \neq 0$

T Continuity of Composition:

• Let $f : I \to \text{image}(f)$ and $g : \text{image}(f) \to \mathbb{R}$ be continuous. Then the composition

$$g \circ f$$

is continuous and

$$\lim_{x \to x_0} g(f(x)) = g\left(\lim_{x \to x_0} f(x)\right).$$

T Continuity of the Inverse:

• Let I be an interval and let $f : I \to J$ be continuous and bijective. Then J is also an interval and the inverse function

$$f^{-1} : J \to I$$

is continuous.

T Intermediate Value Theorem:

• Let $f : [a, b] \to \mathbb{R}$ be continuous and let c be a value between $f(a)$ and $f(b)$. Then there exists $x \in [a, b]$ such that

$$f(x) = c.$$

C Zero of a Continuous Function (Bolzano theorem):

• Let $f : I \subseteq \mathbb{R} \to \mathbb{R}$ be continuous and suppose

$$f(a) < 0, \quad f(b) > 0.$$

Then there exists $x \in (a, b)$ such that

$$f(x) = 0.$$

D Compact Interval:

• A bounded and closed interval $[a, b]$ is called *compact*.

L Bolzano–Weierstrass for Intervals:

• Let (x_n) be a sequence in a compact interval $[a, b]$. Then there exists a subsequence (x_{n_k}) such that

$$x_{n_k} \to x_0 \quad \text{with } x_0 \in [a, b].$$

D Local Extrema:

• Let $f : D \subseteq \mathbb{R} \to \mathbb{R}$.

- x_0 is a *local maximum* if

$$f(x_0) \geq f(x) \quad \forall x \in D.$$

- x_0 is a *local minimum* if

$$f(x_0) \leq f(x) \quad \forall x \in D.$$

- Local maxima and minima are called *local extrema*.

T Extreme Value Theorem:

• Let $f : I \subseteq \mathbb{R} \to \mathbb{R}$ be continuous and let I be compact. Then f is bounded and attains its maximum and minimum on I, i.e., there exist $x_{\min}, x_{\max} \in I$ such that

$$f(x_{\min}) \leq f(x) \leq f(x_{\max}) \quad \forall x \in I.$$

5 Derivatives

5.1 Differentiation

D Derivative:

• Let $f : D \to \mathbb{R}$ and assume D has no isolated points. The *difference quotient* is

$$\frac{f(a+h)-f(a)}{h}.$$

If the limit exists, the *derivative* of f at a is defined by

$$f'(a) = \lim_{h\to 0}\frac{f(a+h)-f(a)}{h} = \lim_{b\to a}\frac{f(b)-f(a)}{b-a}.$$

Interpretation:

- instantaneous rate of change
- slope of the tangent
- sign indicates increase/decrease

D Derivative Function:

• Given $f : D \to \mathbb{R}$, the *derivative function* is

$$x \mapsto f'(x).$$

D Differentiability:

• A function f is *differentiable at* x if $f'(x)$ exists.If this holds for all x in the domain, f is called differentiable.

L Differentiability Implies Continuity:

• If f is differentiable at x_0, then f is continuous at x_0.

D One-Sided Derivatives:

•

- Right derivative:

$$\lim_{h\to 0^+}\frac{f(a+h)-f(a)}{h}$$

- Left derivative:

$$\lim_{h\to 0^-}\frac{f(a+h)-f(a)}{h}$$

D Higher Derivatives:

• Define iteratively:

$$f^{(0)} = f, \quad f^{(1)} = f', \quad f^{(2)} = f'', \quad f^{(n+1)} = (f^{(n)})'.$$

If $f^{(n)}$ exists, f is n-times differentiable.If all derivatives up to order n are continuous, then $f \in C^n(D)$.Define

$$C^\infty(D) := \bigcap_{n=0}^{\infty} C^n(D),$$

called *smooth functions.*

T Basic Derivatives:

•

- Power:

$$(ax^p)' = apx^{p-1}$$

- Exponential:

$$(a^x)' = \ln(a)\, a^x$$

- Trigonometric:

$$(\sin x)' = \cos x, \quad (\cos x)' = -\sin x, \quad (\tan x)' = \frac{1}{\cos^2 x}$$

- Hyperbolic:

$$(\sinh x)' = \cosh x, \quad (\cosh x)' = \sinh x$$

- Inverse:

$$(\ln x)' = \frac{1}{x}, \quad (\arctan x)' = \frac{1}{1+x^2}$$

T Sum and Product Rule:

• For differentiable f, g:

$$(f+g)' = f' + g'$$

$$(fg)' = f'g + fg'$$

T Chain Rule:

• If f is differentiable at x_0 and g at $f(x_0)$, then

$$(g \circ f)'(x_0) = g'(f(x_0)) \cdot f'(x_0).$$

T Quotient Rule:

• If $g(x_0) \neq 0$, then

$$\left(\frac{f}{g}\right)'(x_0) = \frac{f'(x_0)g(x_0) - f(x_0)g'(x_0)}{g(x_0)^2}.$$

T Derivative of the Inverse:

• Let f be bijective and differentiable with $f'(x_0) \neq 0$.Then

$$(f^{-1})'(f(x_0)) = \frac{1}{f'(x_0)}.$$

T Necessary Condition for Extremum:

• If x_0 is a local extremum and f is differentiable at x_0, then

$$f'(x_0) = 0.$$

T Characterization of Local Extrema:

• If x_0 is a local extremum, then at least one holds:

- x_0 is a boundary point
- f is not differentiable at x_0
- $f'(x_0) = 0$

T Rolle's Theorem:

• Let f be continuous on $[a, b]$ and differentiable on (a, b) with $f(a) = f(b)$.Then there exists $\xi \in (a, b)$ such that

$$f'(\xi) = 0.$$

T Mean Value Theorem:

• Let f be continuous on $[a, b]$ and differentiable on (a, b).Then there exists $\xi \in (a, b)$ such that

$$f'(\xi) = \frac{f(b)-f(a)}{b-a}.$$

T l'Hospital's Rule (0/0):

• If

$$\lim_{x\to a} f(x) = \lim_{x\to a} g(x) = 0$$

and

$$\lim_{x\to a}\frac{f'(x)}{g'(x)} = L,$$

then

$$\lim_{x\to a}\frac{f(x)}{g(x)} = L.$$

T l'Hospital's Rule (∞/∞):

• If

$$\lim_{x\to a} |f(x)| = \lim_{x\to a} |g(x)| = \infty$$

and

$$\lim_{x\to a}\frac{f'(x)}{g'(x)} = L,$$

then

$$\lim_{x\to a}\frac{f(x)}{g(x)} = L.$$

T l'Hospital at Infinity:

• If

$$\lim_{x\to\infty} f(x), g(x) \in \{0, \infty\}$$

and

$$\lim_{x\to\infty}\frac{f'(x)}{g'(x)} = L,$$

then

$$\lim_{x\to\infty}\frac{f(x)}{g(x)} = L.$$

T Monotonicity via Derivative:

•

$$f'(x) \geq 0 \iff f \text{ is increasing}$$

C Constant Functions:

• f is constant iff

$$f'(x) = 0 \quad \forall x.$$

T Convexity via First Derivative:

• f is convex iff f' is increasing.

C Convexity via Second Derivative:

• If f is twice differentiable:

$$f''(x) \geq 0 \iff f \text{ is convex.}$$

$$f''(x) \leq 0 \iff f \text{ is concave}$$

T Second Derivative Test:

• Let f be twice differentiable near x_0 and suppose $f'(x_0) = 0$.
Then: $f''(x_0) > 0 \quad \Rightarrow \quad x_0$ is a local minimum,
and $f''(x_0) < 0 \quad \Rightarrow \quad x_0$ is a local maximum.

5.2 Taylor Approximation and Power Series

Local Approximation by Polynomials:

• In a sufficiently small neighborhood of a point x_0, the graph of a differentiable function f behaves almost like its tangent at x_0. Hence, the function can locally be approximated by its tangent.

D Linear Approximation / Tangent Approximation:

• If f is differentiable at x_0, then

$$f(x) \approx f(x_0) + f'(x_0)(x - x_0).$$

The expression

$$P_1(x) = f(x_0) + f'(x_0)(x - x_0)$$

is called the *linearization*, *tangent approximation*, or *first Taylor polynomial* of f at x_0.

T Taylor's Theorem:

• Let f be infinitely differentiable on an open interval I containing x_0. Then for every $x \in I$,

$$f(x) = f(x_0) + f'(x_0)(x - x_0) + \cdots + \frac{1}{n!} f^{(n)}(x_0)(x - x_0)^n,$$

for some c between x_0 and x. Equivalently,

$$f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k + R_n(x),$$

where

$$R_n(x) = \frac{1}{(n+1)!} f^{(n+1)}(c)(x - x_0)^{n+1}.$$

The polynomial

$$P_n(x) = \sum_{k=0}^{n} \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k$$

is called the *n-th Taylor polynomial.*

Taylor Approximation and Power Series:

• Taylor polynomials provide increasingly accurate approximations of many functions. For many functions, taking infinitely many terms recovers the original function.

D Taylor Series:

• For a function f, the series

$$\sum_{k=0}^{\infty} \frac{f^{(k)}(a)}{k!}(x - a)^k$$

is called the *Taylor series* (or Taylor expansion) of f around the point a.

T Important Taylor Series:

• The following series converge for all x:

$$\sin(x) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots$$

$$\cos(x) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots$$

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$$

T Further Important Power Series:

• The following series converge for $-1 < x < 1$.

$$\ln(1 + x) = \sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^n}{n} = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots$$

$$\frac{1}{1 - x} = \sum_{n=0}^{\infty} x^n = 1 + x + x^2 + x^3 + \cdots$$

$$(1 + x)^p = \sum_{n=0}^{\infty} \binom{p}{n} x^n = 1 + px + \frac{p(p-1)}{2!} x^2 + \cdots$$

Operations on Taylor Series:

• New Taylor series can be constructed from known ones using:

- substitution
- multiplication
- differentiation
- integration

T Multiplication of Power Series:

• Let

$$f(x) = \sum_{n=0}^{\infty} a_n x^n, \qquad g(x) = \sum_{n=0}^{\infty} b_n x^n,$$

with x inside both convergence intervals. Then

$$f(x)g(x) = \sum_{n=0}^{\infty} \left(\sum_{k=0}^{n} a_k b_{n-k} \right) x^n.$$

T Termwise Differentiation:

• Let

$$f(x) = \sum_{n=0}^{\infty} a_n x^n$$

inside its convergence interval. Then f is differentiable and

$$f'(x) = \sum_{n=1}^{\infty} n a_n x^{n-1}.$$

T Termwise Integration:

• Let

$$f(x) = \sum_{n=0}^{\infty} a_n x^n$$

inside its convergence interval I. Then for every interval $[a, b] \subseteq I$,

$$\int_a^b f(x)\, dx = \sum_{n=0}^{\infty} \int_a^b a_n x^n \, dx.$$

6 Differential equations

D Differential Equation:

• A *differential equation* is an equation involving an unknown function together with its derivatives.

$$u'(t) = r(a - u(t))(b - u(t))$$

where r, a, b are constants, another example

$$y^{(4)}(x) - 4y^{(3)}(x) + 5y''(x) - 4y'(x) + 4y(x) = 0.$$

The variables t, x are called *independent variables*, while u, y are called *dependent variables.*

D Linear Differential Equation:

• A differential equation of the form

$$a_n(x)y^{(n)}(x) + a_{n-1}(x)y^{(n-1)}(x) + \cdots + a_1(x)y'(x) + a_0(x)y(x) = s(x)$$

is called a *linear differential equation* with coefficients $a_i(x)$.

The *order* of the differential equation is the highest order derivative appearing in the equation. The function $s(x)$ is called the *forcing term.* If $s(x) = 0$, the equation is called *homogeneous*; otherwise it is called *inhomogeneous.*

D Differential Equation with Constant Coefficients:

• If all coefficient functions $a_i(x)$ are constant functions, the equation is called a *differential equation with constant coefficients.*

T Superposition Principle:

• If $y_1(x)$ and $y_2(x)$ are solutions of a linear homogeneous differential equation, then

$$y(x) = C_1y_1(x) + C_2y_2(x), \qquad C_1, C_2 \in \mathbb{R},$$

is also a solution of the same differential equation.

T Fundamental Solutions:

• A linear homogeneous differential equation of order n possesses n solutions $y_1(x), \dots, y_n(x)$ such that the general solution is given by

$$y(x) = C_1y_1(x) + \cdots + C_ny_n(x), \qquad C_1, \dots, C_n \in \mathbb{R}.$$

These solutions are called *fundamental solutions.*

Characteristic Polynomial:

• Consider the differential equation

$$a_nu^{(n)}(x) + a_{n-1}u^{(n-1)}(x) + \cdots + a_1u'(x) + a_0u(x) = 0.$$

Using the ansatz $u(x) = e^{\lambda x}$ (Euler ansatz), we obtain the *characteristic equation*

$$a_n\lambda^n + a_{n-1}\lambda^{n-1} + \cdots + a_1\lambda + a_0 = 0.$$

We get the *characteristic polynomial*:

$$p(\lambda) = a_n\lambda^n + a_{n-1}\lambda^{n-1} + \cdots + a_1\lambda + a_0$$

T Solution of Linear Homogeneous Differential Equations:

• Consider the differential equation

$$a_nu^{(n)}(x) + a_{n-1}u^{(n-1)}(x) + \cdots + a_1u'(x) + a_0u(x) = 0, \qquad a_i \in \mathbb{R}.$$

If all roots λ_i of the characteristic polynomial are real and have multiplicities m_i, then $u(x) = \sum_{i=1}^{r}\left(\sum_{p=0}^{m_i-1} C_{ip}x^pe^{\lambda_i x}\right)$.
If there are additionally complex conjugate roots $a_j \pm ib_j$ with multiplicities m_j, then $u(x) = \sum_{i=1}^{r}\left(\sum_{p=0}^{m_i-1} C_{ip}x^pe^{\lambda_i x}\right) + \sum_{j=1}^{s}\left(\sum_{q=0}^{m_j-1} A_{jq}x^qe^{a_jx}\sin(b_jx) + B_{jq}x^qe^{a_jx}\cos(b_jx)\right)$.

Constants in the General Solution:

• The general solution of a differential equation of order n contains n constants. Therefore, in order to determine a unique solution, one must specify n additional conditions.

D Boundary Value Problem:

• Conditions imposed at different points, e.g. for $n = 2$:
$u(t_0) = u_0, \qquad u(t_1) = u_1, \qquad t_0 \neq t_1,$
define a *boundary value problem.*

D Initial Value Problem:

• Conditions imposed at a single point, e.g. for $n = 2$:
$u(t_0) = u_0, \qquad u'(t_0) = v_0,$
define an *initial value problem.*

7 Integration

8 Integration

D Antiderivative / Indefinite Integral:

• Let $I \subseteq \mathbb{R}$ be an interval and let

$$f : I \to \mathbb{R}$$

be a function. A differentiable function

$$F : I \to \mathbb{R}$$

satisfying

$$F'(x) = f(x) \quad \forall x \in I$$

is called an *antiderivative* of f on I. Determining F from f is called *indefinite integration*.

Non-Uniqueness of Antiderivatives:

• An antiderivative is not unique. If $F(x)$ is an antiderivative of $f(x)$, then

$$F(x) + C$$

is also an antiderivative for every constant $C \in \mathbb{R}$. Hence a function possesses an entire family of antiderivatives. This family is called the *indefinite integral* and is written as

$$\int f(x)\,dx.$$

The function f is called the *integrand*.

T Linearity of the Indefinite Integral:

• For indefinite integrals:

$$\int (f(x) \pm g(x))\,dx = \int f(x)\,dx \pm \int g(x)\,dx$$

and

$$\int s f(x)\,dx = s \int f(x)\,dx,$$

where s is a constant.

T Basic Integration Rules:

• Rules:

- $\int k\,dx = kx + C$
- $\int x^n\,dx = \frac{1}{n+1}x^{n+1} + C, \qquad n \neq -1$
- $\int e^x\,dx = e^x + C$
- $\int \frac{1}{x}\,dx = \ln(|x|) + C$
- $\int \sin(x)\,dx = -\cos(x) + C$
- $\int \cos(x)\,dx = \sin(x) + C$
- $\int \sinh(x)\,dx = \cosh(x) + C$
- $\int \cosh(x)\,dx = \sinh(x) + C$

Differentiation Rules vs. Integration Rules:

• Recall the two fundamental differentiation rules:

$$\frac{d}{dx} f(g(x)) = f'(g(x)) \cdot g'(x)$$

(chain rule) and

$$\frac{d}{dx}(f(x)g(x)) = f'(x)g(x) + f(x)g'(x)$$

(product rule). The goal is to derive and apply the corresponding rules in integration theory.

T Integration by Parts:

• Using the product rule:

$$\frac{d}{dx}(f(x)g(x)) = f'(x)g(x) + f(x)g'(x),$$

we obtain

$$\int \frac{d}{dx}(f(x)g(x))\,dx = (fg)(x).$$

Therefore,

$$(fg)(x) = \int f'(x)g(x)\,dx + \int f(x)g'(x)\,dx.$$

Rearranging gives the formula for *integration by parts*:

$$\boxed{\int f'(x)g(x)\,dx = f(x)g(x) - \int f(x)g'(x)\,dx}$$

T Integration by Substitution:

• Using the chain rule:

$$\frac{d}{dx} f(g(x)) = f'(g(x))g'(x),$$

we obtain

$$\int f'(g(x))g'(x)\,dx = f(g(x)) + C.$$

With the substitution

$$y = g(x),$$

this becomes

$$\int f'(g(x))g'(x)\,dx = \int \frac{df}{dy}\,dy.$$

Hence the substitution rule is

$$\boxed{\int f'(g(x))g'(x)\,dx = f(g(x)) + C}$$

9 Integration Part 2

Motivation:

• Given a velocity function $v(t)$ describing one-dimensional motion, natural questions are:

- What is the total distance traveled?
- What is the change in position?

D Partition of an Interval:

• Let $[a, b] \subseteq \mathbb{R}$ be a compact interval. A *partition* of $[a, b]$ is a finite set of points

$$a = x_0 < x_1 < \cdots < x_{n-1} < x_n = b.$$

The points x_i are called *subdivision points*. A partition

$$a = y_0 < y_1 < \cdots < y_m = b$$

is called a *refinement* of

$$a = x_0 < x_1 < \cdots < x_n = b$$

if

$$\{x_0, \ldots, x_n\} \subseteq \{y_0, \ldots, y_m\}.$$

D Step Function:

• A function

$$f : [a, b] \to \mathbb{R}$$

is called a *step function* if there exists a partition

$$a = x_0 < x_1 < \cdots < x_n = b$$

such that for every $k = 1, \ldots, n$, the restriction

$$f|_{(x_{k-1}, x_k)}$$

is constant.

D Integral of a Step Function:

• Let

$$f : [a, b] \to \mathbb{R}$$

be a step function with respect to the partition

$$a = x_0 < x_1 < \cdots < x_n = b.$$

Then the integral of f over $[a, b]$ is defined by

$$\int_a^b f(x)\,dx := \sum_{k=1}^{n} c_k (x_k - x_{k-1}),$$

where c_k denotes the constant value of f on (x_{k-1}, x_k).

D Upper and Lower Sums:

• Let

$$f : [a, b] \to \mathbb{R}.$$

The set of upper sums is defined by

$$U(f) := \left\{ \int_a^b s(x)\,dx \;\middle|\; s \in TF,\ s \geq f \right\},$$

and the set of lower sums by

$$L(f) := \left\{ \int_a^b s(x)\,dx \;\middle|\; s \in TF,\ s \leq f \right\},$$

where TF denotes the set of step functions.

D Riemann Integral:

- A bounded function

$$f : [a,b] \to \mathbb{R}$$

is called *Riemann integrable* if

$$\sup L(f) = \inf U(f).$$

In this case, the common value is called the *Riemann integral* and is denoted by

$$\int_a^b f(x)\,dx := \sup L(f) = \inf U(f).$$

The numbers a, b are called the lower and upper limits of integration, and f is called the integrand.

T Properties of the Riemann Integral:

-
  (i) (Linearity) If $f, g : [a,b] \to \mathbb{R}$ are integrable and $\alpha, \beta \in \mathbb{R}$, then

  $$\int_a^b (\alpha f + \beta g)(x)\,dx = \alpha \int_a^b f(x)\,dx + \beta \int_a^b g(x)\,dx.$$

  (ii) (Monotonicity) If

  $$f(x) \leq g(x)$$

  for all x, then

  $$\int_a^b f(x)\,dx \leq \int_a^b g(x)\,dx.$$

  (iii) (Triangle Inequality)

  $$\left| \int_a^b f(x)\,dx \right| \leq \int_a^b |f(x)|\,dx.$$

  (iv) (Reversal of Integration Limits)

  $$\int_a^b f(x)\,dx = -\int_b^a f(x)\,dx.$$

  (v) (Splitting the Interval) For $a \leq c \leq b$:

  $$\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx.$$

T Monotone Functions are Integrable:

- Every monotone function

$$f : [a,b] \to \mathbb{R}$$

is Riemann integrable.

T Continuous Functions are Integrable:

- Every continuous function

$$f : [a,b] \to \mathbb{R}$$

is Riemann integrable.

D Pointwise Convergence of Functions:

- Let

$$(f_n)_{n \in \mathbb{N}_0}$$

be a sequence of functions

$$f_n : D \to \mathbb{R},$$

and let

$$f : D \to \mathbb{R}.$$

The sequence (f_n) converges *pointwise* to f if for every $x \in D$, the sequence

$$(f_n(x))_{n \in \mathbb{N}_0}$$

converges to $f(x)$.

D Uniform Convergence:

- The sequence (f_n) converges *uniformly* to f if

$$\forall \varepsilon > 0\ \exists N$$

such that for all

$$n \geq N \quad \text{and all } x \in D,$$

we have

$$|f_n(x) - f(x)| < \varepsilon.$$

T Uniform Limit of Continuous Functions:

- Let

$$(f_n)_{n \in \mathbb{N}_0}$$

be a sequence of continuous functions

$$f_n : D \to \mathbb{R}$$

converging uniformly to

$$f : D \to \mathbb{R}.$$

Then f is continuous.

T Uniform Limit and Integration:

- Let

$$(f_n)_{n \in \mathbb{N}_0}$$

be a sequence of integrable functions

$$f_n : [a,b] \to \mathbb{R}$$

converging uniformly to

$$f : [a,b] \to \mathbb{R}.$$

Then f is integrable and

$$\int_a^b f(x)\,dx = \lim_{n\to\infty} \int_a^b f_n(x)\,dx.$$

T Mean Value Theorem for Integrals:

- If

$$f : [a,b] \to \mathbb{R}$$

is continuous, then there exists

$$c \in [a,b]$$

such that

$$f(c) = \frac{1}{b-a} \int_a^b f(x)\,dx.$$

The quantity

$$\frac{1}{b-a} \int_a^b f(x)\,dx$$

is called the *average value* of f on $[a,b]$.

T Fundamental Theorem of Calculus:

- Let

$$f : [a,b] \to \mathbb{R}$$

be continuous. Then for every constant $C \in \mathbb{R}$, the function

$$F(x) := \int_a^x f(y)\,dy + C$$

is an antiderivative of f. Moreover, every antiderivative of f is of this form.

C Integral Representation:

- Let

$$F : [a,b] \to \mathbb{R}$$

be continuously differentiable. Then

$$F(x) = F(a) + \int_a^x F'(t)\,dt.$$

C Newton–Leibniz Formula:

- Let

$$f : [a,b] \to \mathbb{R}$$

be continuous and let F be an antiderivative of f. Then

$$\int_a^b f(t)\,dt = F(b) - F(a).$$

T Integration by Parts:

- Let

$$f, g : [a, b] \to \mathbb{R}$$

be continuously differentiable. Then

$$\int_a^b f(x)g'(x)\,dx = f(x)g(x)\Big|_a^b - \int_a^b f'(x)g(x)\,dx.$$

T Substitution Rule (Version 1):

- Let

$$f : I \to J$$

be continuously differentiable and

$$g : J \to \mathbb{R}$$

continuous. Then for every

$$[a, b] \subseteq I,$$

$$\int_a^b g(f(x))f'(x)\,dx = \int_{f(a)}^{f(b)} g(y)\,dy.$$

T Substitution Rule (Version 2):

- Let

$$f : I \to J$$

be continuously differentiable and

$$g : J \to \mathbb{R}$$

continuous. Assume

$$f'(x) \neq 0 \quad \forall x \in [a, b]$$

and let

$$f^{-1} : [f(a), f(b)] \to \mathbb{R}$$

denote the inverse of $f|_{[a,b]}$. Then

$$\int_a^b g(f(x))\,dx = \int_{f(a)}^{f(b)} g(y)(f^{-1})'(y)\,dy.$$

$\lim_{x\to\infty} \frac{1}{x} = 0$	$\lim_{x\to\infty} \left(1+\frac{1}{x}\right) = 1$	$\lim_{x\to 0} \frac{\log(1-x)}{x} = -1$	$\lim_{x\to 0} x \log x = 0$
$\lim_{x\to\infty} e^x = \infty$	$\lim_{x\to -\infty} e^x = 0$	$\lim_{x\to 0} \frac{1-\cos x}{x^2} = \frac{1}{2}$	$\lim_{x\to 0} \frac{e^x - 1}{x} = 1$
$\lim_{x\to\infty} e^{-x} = 0$	$\lim_{x\to -\infty} e^{-x} = \infty$	$\lim_{x\to 0} \frac{x}{\arctan x} = 1$	$\lim_{x\to\infty} \arctan x = \frac{\pi}{2}$
$\lim_{x\to\infty} \frac{e^x}{x^m} = \infty$	$\lim_{x\to -\infty} x e^x = 0$	$\lim_{x\to\infty} \left(\frac{x}{x+k}\right)^x = e^{-k}$	$\lim_{x\to 0} \frac{e^x - 1}{x} = 1$
$\lim_{x\to\infty} \ln x = \infty$	$\lim_{x\to 0} \ln x = -\infty$	$\lim_{x\to 0} \frac{a^x - 1}{x} = \ln(a),\ a > 0$	$\lim_{x\to 0} \frac{e^{ax} - 1}{x} = a$
$\lim_{x\to\infty} (1+x)^{1/x} = 1$	$\lim_{x\to 0} (1+x)^{1/x} = e$	$\lim_{x\to 0} \frac{\ln(1+x)}{x} = 1$	$\lim_{x\to 1} \frac{\ln x}{x-1} = 1$
$\lim_{x\to\infty} \left(1+\frac{1}{x}\right)^b = 1$	$\lim_{x\to\infty} \left(1+\frac{1}{x}\right)^b = 1$	$\lim_{x\to\infty} \frac{\ln x}{x} = 0$	$\lim_{x\to\infty} \frac{\log x}{x^\alpha} = 0$
$\lim_{x\to\infty} x^a q^x = 0,\ 0 \leq q < 1$	$\lim_{x\to\infty} n^{1/n} = 1$	$\lim_{x\to\infty} \sqrt[x]{x} = 1$	$\lim_{x\to\infty} \frac{2x}{2^x} = 0$
$\lim_{x\to\pm\infty} \left(1+\frac{1}{x}\right)^x = e$	$\lim_{x\to\infty} \left(1-\frac{1}{x}\right)^x = \frac{1}{e}$	$\lim_{x\to(\pi/2)^-} \tan x = +\infty$	$\lim_{x\to(\pi/2)^+} \tan x = -\infty$
$\lim_{x\to\pm\infty} \left(1+\frac{k}{x}\right)^{mx} = e^{km}$	$\lim_{x\to 0} \frac{\sin x}{x} = 1$	$\lim_{x\to\infty} \frac{\sin x}{x} = 0$	$\lim_{x\to 0^+} x \ln x = 0$
$\lim_{x\to 0} \frac{1}{\cos x} = 1$	$\lim_{x\to 0} \frac{\cos x - 1}{x} = 0$		

Table 1: Important limits

$f(x)$		$g(x)$		$h(x)$	
$f(x)$	$f'(x)$	$g(x)$	$g'(x)$	$h(x)$	$h'(x)$
$\sin(x)$	$\cos(x)$	$\tan(x)$	$\frac{1}{\cos^2(x)}$	$\arcsin(x)$	$\frac{1}{\sqrt{1-x^2}}$
$\cot(x)$	$-\frac{1}{\sin^2(x)}$	$mx+b$	m	$\arccos(x)$	$-\frac{1}{\sqrt{1-x^2}}$
$\cos(x)$	$-\sin(x)$	$r \cdot g(x)$	$r \cdot g'(x)$	$\arctan(x)$	$\frac{1}{1+x^2}$
x^n	nx^{n-1}	e^x	e^x	$\operatorname{arccot}(x)$	$-\frac{1}{1+x^2}$
$\ln(x)$	$\frac{1}{x}$	a^x	$a^x \ln(a)$	$\operatorname{arsinh}(x)$	$\frac{1}{\sqrt{1+x^2}}$
$\log_a(x)$	$\frac{1}{x\ln(a)}$	$\sinh(x)$	$\cosh(x)$	$\operatorname{arcosh}(x)$	$\frac{1}{\sqrt{x^2-1}}$
$\tanh(x)$	$\frac{1}{\cosh^2(x)}$	$\cosh(x)$	$\sinh(x)$	$\operatorname{artanh}(x)$	$\frac{1}{1-x^2}$
$\tan(x)$	$1+\tan^2(x)$				

Table 2: Classic derivatives

Nr.	Regel
1	$(f+g)'(x) = f'(x) + g'(x)$
2	$(f \cdot g)'(x) = f'(x)\, g(x) + f(x)\, g'(x)$
3	$(g \circ f)'(x) = g'(f(x)) \cdot f'(x)$
4	$\left(\frac{f}{g}\right)'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{g(x)^2}$
5	$(f^{-1})'(y_0) = \frac{1}{f'(x_0)}, \quad \text{mit } y_0 = f(x_0)$
Note: This holds only if f^{-1} exists and f is continuous.	

Table 3: Calculation rules of derivatives